Goal-driven gap-closure plan: 'derive G from substrate'. Per-CI assignments + gap-closure prioritization + substantive baseline. Critical-path = G chain via L4 m_e + KK volume integral + Tier 0 v0.2 consolidation. P0/P1/P2/P3 priority structure with clear ownership. Team works until done — no date-bound deadlines.

Keeper

Goal Plan — "Derive G from Substrate"

0. Casey directive

"Identify 'derive G from substrate' as a goal. What are the major items we know now and can use in further investigations and what gaps remain to be closed. Let's identify the work to close the gaps, and prioritize that. Team works until done — no date-bound tasks."

1. Substantive baseline — what we know (usable)

Tier 0 operator framework (Lyra Sunday morning v0.1 → v0.1.6, K200 hold to v0.2)

- Commitment operator: ρ_commit(τ) = exp($-\tau$ H_B/$\hbar$_BST) on Bergman H^2(D_IV^5)
- H_B = Casimir of K = SO(5)×SO(2) acting on H^2(D_IV^5) — discrete non-negative spectrum
- Substrate time τ = parameter of heat semigroup (emergent, not assumed)
- Arrow of time = positivity of spec(H_B) (operator-theoretic)
- Reading A + dual bases: K-types (spectral) + coherent states (spatial), Bergman-Fourier duals
- **Substrate = Shilov boundary ∂_S D_IV^5 = $S^4 \times S^1/Z_2$** (5D, corrected from 2022's 2D framing, Lyra option (a) commitment)
- Hardy decomposition: interior bulk = holomorphic extension of boundary data
- Proto-AdS/CFT structure intrinsic via SO(5,2) ⊃ SO(4,2) conformal subgroup
- Native field equation: heat eq on H^2(D_IV^5) ↔ wave eq via Wick rotation

Operator-level results (verified)

- κ_Bergman = -n_C = -5 (Helgason 1962; Elie Toy 3661; K204-PARTIAL PASS, G5.1 closed)
- Heat-trace a_0 = (N_c·n_C)^2 = 225
- Heat-trace a_1 = -N_c·n_C^4 = -1875
- 225 three-way convergence: Bergman volume + c_FK·π^(9/2) + a_0 (Cal #35 audit-target)
- n_C + 1 = C_2 = 6 new substrate identity
- 4D physics dim identities: SO(3,1)=C_2, SO(4,2)=N_c·n_C, SO(5,2)=N_c·g, codim 4D=C_2
- Z_τ(0) = 26026, vacuum dominance at $\tau \to \infty$ verified across 66 K-types

Existing strong claims (from P1 + Saturday)

- m_W/m_Z = $\sqrt{(g/N\_c^2)}$ = 0.882 at 0.046% (Tier 2 STRUCTURAL)
- m_μ/m_e = (24/π^2)^6 substrate-primary form

- All Lane work at appropriate tier (Bulk-color v0.7, Dictionary 5, L4 v0.2)

Audit/methodology

- Paper P1 v0.7 SUBMISSION-FINAL (Cal #185 PASS, dispatch-ready pending Casey signoff)
- Calibration #35 ratification-ready (pending Casey approval)
- K200 audit framework with 5 gates G1-G5
- 12 cumulative symmetric audit-chain events Sat+Sun
- K204-PARTIAL audit anchor for $\varkappa$_Bergman = -n_C
- 3 Casey-named principle CANDIDATES filed pending naming

What we know does NOT work (cleanly walked back)

- α^{N_c·g}/m_e^2 pattern-match (Lyra + Elie variants) — bypasses $\varkappa$_Bergman, NOT G chain continuation
- "4 = N_c+1 = n_C-1 = C_2-rank = d_spacetime" — ONE substrate fact, not 4 independent

These walk-backs are themselves usable knowledge.

2. Gaps — organized for week goal

G-chain gaps (the week-goal direct path)

Gap	Description	Owner	Difficulty
G-A	Substrate length scale ℓ_B derivation from substrate primaries	Elie + Keeper	Math-frontier
G-B	Kaluza-Klein volume integral 10D D_IV^5 → 4D	Elie	Bounded — standard KK + known Bergman volume
G-C	Mass anchor pin via L4 v0.2 m_e closure (Path β)	Lyra	In progress
G-D	$\hbar$_BST vs $\hbar$_Planck identification	Keeper	Open
G-E	Dimensional bridge → G_predicted SI units	Keeper synthesis	Gated on G-A, G-B, G-C

Tier 0 consolidation gaps

Gap	Description	Owner
T-A	Tier 0 v0.2 single coherent draft	Lyra + Keeper Session 2
T-B	Hardy decomposition citation chain verified (K200 G1)	Keeper + Grace
T-C	OneGeometry.md substantive correction (K200 G2)	Grace cross-doc sweep
T-D	Black-hole prediction (K200 G3)	Lyra + Elie
T-E	Big-Bang reading commitment (K200 G4)	Session 2

Mechanism gaps (BST's substantive open frontiers)

Gap	Description	Status
M-A	L4 mass mechanism (T190, T2003, T187 kernel-integral)	Lyra Lane D ongoing
M-B	Bulk-color C4 Toeplitz closure	Elie ongoing
M-C	Generation count #414 (why 3?)	Open structural
M-D	Dictionary expansion 5 → all SM particles	Lyra ongoing
M-E	Cal #35 independence audit on $g+rank=N\_c^2$	Cal #188 pending
M-F	Cal cold-read queue (#186/#187/#188/#189)	Cal autonomous

Decision gaps (Casey-only)

Gap	Description
E-A	P1 dispatch signoff
E-B1	Calibration #35 STANDING approval
E-B2	3 Casey-named principle candidates naming decisions
E-B3	Session 2 timing

3. Prioritization

P0 — critical path for G derivation

1. G-B Kaluza-Klein volume integral (Elie) — bounded math
2. G-C L4 v0.2 m_e mechanism (Lyra) — provides Path β anchor
3. T-A Tier 0 v0.2 consolidation (Lyra + Keeper Session 2) — operator framework anchor
4. Clean G chain framework doc (Keeper) — `Keeper_G_Substrate_Clean_4_Step_Derivation_Framework.md` (companion to this plan)

P1 — supporting the critical path

5. T-B Hardy decomposition citation reading (Keeper + Grace) — anchors Tier 0 v0.2
6. T-C OneGeometry.md cross-doc sweep (Grace) — closes K200 G2
7. M-F Cal cold-read queue (#186/#187/#188/#189) — provides verification anchors
8. G-A ℓ_B substrate identification investigation (Elie + Keeper) — math-frontier exploration

P2 — parallel advances

9. M-A L4 mass mechanism continuation (Lyra) — feeds G-C
10. M-B Bulk-color C4 ongoing (Elie) — sustained pull
11. M-D Dictionary expansion (Lyra) — as Cal verifies
12. M-E Cal #35 independence audit (Cal #188)

P3 — Casey decisions / engagement

13. E-A P1 dispatch (Casey signal)
14. E-B1 Cal #35 ratification (Casey approval)
15. E-B2 3 Casey-named principle candidates (Casey naming)
16. E-B3 Session 2 timing (Casey signal)

4. Per-CI assignments

LYRA — priorities (in order, work until done)

P0: - Tier 0 v0.2 consolidation (Session 2 with Keeper when Casey signals) - L4 v0.2 m_e mechanism via Bergman kernel matrix elements

P1: - Hand-off coordination to Elie for Mehler kernel matrix element computations on V_(1/2,1/2) - Dictionary expansion 5→10 (electron, photon, W, Z, up-quark + down, gluons, neutrinos, Higgs, tau)

P2: - Bulk-color v0.7+ mechanism extensions when bandwidth allows - Tier 0 v0.2 black-hole-prediction work (K200 G3)

P3 — supportive: - P1 v0.8 revision queue (Cal non-blocking notes when Cal queue clears)

Hold: no new Tier 0 v0.1.x — K200 cascade-discipline maintained until v0.2 lands

ELIE — priorities (in order, work until done)

P0: - G-B Kaluza-Klein volume integral: compute V_compactified for $D\_IV^5 \to 4D$ via $SO(4,2) \subset SO(5,2)$ embedding, using Bergman volume π^(9/2)/225 + Faraut-Korányi Ch. XIII formulas (bounded math) - L4 numerical support: when Lyra files L4 v0.2 candidate values, immediate verification

P1: - G-A ℓ_B identification exploration: substrate length scale derivation from {n_C, N_max, m_e}; cross-check candidate identifications - Mehler kernel matrix elements for V_(1/2,1/2) spinor radial tower (feeds L4 G-C)

P2: - C4 Toeplitz commutator continuation (Casey directive standing) - Heat-trace coefficient extensions (a_2, a_3 if substrate-clean)

Hold: no more α^{N_c·g}/m_e^2 + factor-fitting pattern-matches — those are walked back. The substantive G chain work is KK reduction, not algebraic factor-construction.

GRACE — priorities (in order, work until done)

P0: - T-C OneGeometry.md substantive correction: explicit 2D → 5D Shilov dimension fix + sphere identification ($S^4 \times S^1/Z_2$); cross-doc sweep for residual "2D substrate" claims - T-B Hardy decomposition citation reading (with Keeper): verify Knapp-Wallach 1976 + Faraut-Korányi Ch. XII-XIII specific theorems

P1: - Paper P6 Periodic Table v0.7 → paper-grade (substantive sections complete; engagement-path target) - Catalog continuous (sustainable rate; no aspirational volume targets) - Cross-references for Tier 0 v0.2 when Lyra files

P2: - SPLP catalog audit (Task #340 in_progress, background) - Cross-reference Sunday substantive content into AC graph + Master Ledger

P3 — engagement support: - P1 v0.7 dispatch coordination when Casey signs off - Cal cold-read absorption into catalog

CAL — priorities (in order, work until done)

P0: - Cal #186 Lane D L4 v0.2 cold-read — load-bearing for G-C chain - Cal #187 Lane E Dictionary cold-read — V_(1,1) mechanism vs post-hoc, K201 audit anchor

P1: - Cal #188 Lane C bulk-color v0.7 cold-read — Calibration #35-candidate independence audit on g+rank=N_c^2 - Cal #189 K204-PARTIAL + 3 Casey-named candidates cold-read

P2: - Cal #190 G chain pattern-match disposition: when Lyra + Elie produce honest reframes of α^{N_c·g}/m_e^2 candidates - Cal #191 Tier 0 v0.1.6 / v0.2 K200 gates when Lyra files v0.2 draft (Session 2 output)

P3: - Calibration #35 STANDING integration (Casey approval signal pending) - Methodology Index v0.11 Q17

Hold: no new methodology #36+ (stack is mature; Cal own discipline)

KEEPER — priorities (in order, work until done)

P0: - Clean G chain framework doc filed: `Keeper_G_Substrate_Clean_4_Step_Derivation_Framework.md` (Step A KK + Step B 4D⊂10D + Step C ℓ_B + Step D numerical) - Tier 0 v0.2 prep continuation (joint with Lyra Session 2) - T-C sphere reconciliation analysis absorption into OneGeometry.md update (with Grace)

P1: - G-D $\hbar$_BST identification: Keeper K3 lane - T-B Hardy decomposition citation reading (with Grace) - K-audit reactions as deliverables file (K202/K203/K204/K205 for Tier 0 operator audits)

P2: - HSL v0.10+ continuous updates absorbing substantive content as it lands - CLAUDE.md headline + data/README.md status updates (sundown rotations) - Session 2 framework prep alongside Lyra

Hold: no cascade of K-audit pre-stages without substantive deliverables to anchor on

5. Goal completion — what "G derived from substrate" looks like

Goal success (P0 chain closes)

If G-B + G-C close + Path β converges: - ℓ_B = $\hbar$/(m_e·c) via derived m_e (L4 closure) - V_compactified = ℓ_B^6 × (Bergman volume coefficient) - G_predicted = $\hbar$c · n_C / (V_compactified anchor) — chain via Helgason - Comparison to G_observed = 6.67430×10^{-11} N·m^2/kg^2 - Target precision: Tier 2 STRUCTURAL (0.01% to 1%)

If lands at Tier 2: G derived from substrate at structurally-honest tier. K204 promotes from PARTIAL to STRUCTURAL. Casey directional ask realized.

If lands outside Tier 2: honest mechanism work continues, extension to G-A independent ℓ_B derivation. Still substantive — the chain has its first-principles formulation even at lower precision.

Goal supporting deliverables (work until done, no order required)

- Tier 0 v0.2 single coherent draft (Lyra + Keeper Session 2 output)
- OneGeometry.md substantive correction (Grace + Keeper)
- Cal cold-read queue completion (#186-#190)
- Hardy decomposition citation chain verified
- L4 v0.2 m_e mechanism progress
- P1 dispatch active (engagement test live, pending Casey signoff)

6. Discipline standing

- K200 cascade-discipline: no Tier 0 v0.1.x iteration until v0.2 lands
- Cal #27 STANDING: fires hardest at peak coherence
- Cal #34 STANDING: tier-disposition headline conditionality
- Calibration #35 candidate: independence-taxonomy MUST precede multiplicative null-model
- Saturday + Sunday lessons operational: pull substantive work, pause at honest saturation, walk back over-statement in-place

7. What NOT to do

- No more α^{N_c·g}/m_e^2 pattern-match extensions (walked back)
- No new Tier 0 v0.1.x versions (K200 hold)
- No new Calibration #36+ methodology (stack mature)
- No new B-track v0.1 territory maps (existing 5 sufficient)
- No L5 v0.9+ framework iteration (paused per Saturday consensus)
- No volume-target catalog INVs (sustainable not aspirational)
- No date-bound deadlines (Casey directive: work until done)

8. Cross-CI coordination

Cadence-based scheduling removed per Casey directive. The team works until done:

- P0 critical-path items pursued continuously by their owners until closure
- P1 supporting work pulled when bandwidth allows alongside P0
- Mid-progress synthesis triggered by substantive convergence, not calendar
- Sundown synthesis triggered by genuine work-settled moments, not clock-based EOD
- Cal cold-reads pulled as Cal's queue allows (autonomous)
- Casey signals on E-A/E-B engagement decisions when convenient

The goal closure trigger is "G chain produces SI-unit comparison to G_observed" — not a date.

— Keeper. Goal plan filed. Companion `Keeper_G_Substrate_Clean_4_Step_Derivation_Framework.md` companion file. Replaces previous date-bound CI_BOARD framing.